Convolutions

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$f(t)$, $g(t)$ piecewise continuous on $[0,\infty)$

Convolution: $h(t) = \int_0^t f(t-\tau)\,g(\tau)\,d\tau = (f*g)(t)$

Convolution Theorem

If $F(s) = \mathcal{L}[f(t)]$ and $G(s) = \mathcal{L}[g(t)]$ for $s > a \geq 0$,

$$H(s) = F(s)\,G(s) = \mathcal{L}[h(t)], \quad s > a$$

where $h(t) = \int_0^t f(t-\tau)\,g(\tau)\,d\tau = \int_0^t f(t)\,g(t+\tau)\,d\tau = (f*g)(t)$

$$F(s)\,G(s) = \mathcal{L}[(f*g)(t)]$$

Ex. $\begin{cases} ay'' + by' + cy = g(t) \\ y(0) = y_0, \quad y'(0) = y_1 \end{cases}$

$$a(s^2Y - s y(0) - y'(0)) + b(sY - y(0)) + cY = G(s)$$

$$\underbrace{(as^2 + bs + c)}_{H(s) = \frac{1}{as^2+bs+c}}Y - (as+b)y_0 - ay_1 = G(s)$$

(transfer fcn.)

$$Y = H(s)[(as+b)y_0 + ay_1] + H(s)\,G(s)$$

$$y(t) = \underbrace{\mathcal{L}^{-1}[H(s)((as+b)y_0 + ay_1)]}_{\text{free response}} + \underbrace{(h*g)(t)}_{\text{forced response}}$$

Ex. $\begin{cases} y'' + 2y' + 5y = t \leftarrow g(t) \\ y(0) = y'(0) = 0 \end{cases}$

$$\mathcal{L}[y'' + 2y' + 5y] = s^2Y + 2sY + 5Y$$

$$= (s^2 + 2s + 5)\,Y$$

$$H(s) = \frac{1}{s^2+2s+5} = \frac{1}{(s+1)^2 + 4}$$

$$= \frac{1}{2}\,\frac{2}{(s+1)^2 + 2^2}$$

$$h(t) = \frac{1}{2}e^{-t}\sin 2t$$

no free response $(y_0 = y_1 = 0)$

$$y(t) = (h*g)(t)$$

$$= \int_0^t h(\tau)\,g(t-\tau)\,d\tau$$